

Intra-Abdominal Infections – Summary

Key Symptoms: Depends on the type of intra-abdominal infection

- Common to all types: abdominal pain, fever, N/V, jaundice, abnormal lab readings, etc.

Most common causative pathogens:

- **Primary Peritonitis:** E. coli, K. pneumoniae, S. pneumoniae, other Streptococci
- **Secondary Peritonitis:** Enterobacteriales, anaerobes, enterococcus, yeast, multi-drug resistant organisms (MDROs)
- **Cholangitis and Cholecystitis:** E. coli, Klebsiella spp., enterococcus spp., anaerobes
- **Acute Appendicitis:** E. coli, enterococcus spp., anaerobes
- **Abscess:** Gut anaerobes (B. fragilis), gram-negative bacteria, Candida (less common)
- **Acute pancreatitis:** Non-infectious

Non-Pharmacological Treatment:

- **Cholangitis:** Hydration + biliary decompression
- **Cholecystitis:** Elective surgery after improvement in infection
- **Acute appendicitis:** Surgery
- **Abscess:** Surgical debridement and drainage
- **Acute Pancreatitis:** Fluid resuscitation, pain control + nutritional support

Pharmacological Treatment: Empiric therapy options are as follows:

- **Primary Peritonitis:** 3rd gen cephalosporin, fluoroquinolones (FQs), carbapenems
- **Secondary Peritonitis:** 1st/2nd/3rd gen cephalosporin + metronidazole **OR** amoxi-Clav **OR** Pip-Tazo **OR** Carbapenem **OR** Aminoglycosides + metronidazole **or** FQ +/- metronidazole
- **Cholangitis and Cholecystitis:** 2nd/3rd gen cephalosporin +/- metronidazole **OR** amoxi-clav **OR** Pip-Tazo **OR** Carbapenems
- **Acute Appendicitis (infective):** 1st/2nd/3rd gen cephalosporin + metronidazole (no more than 1-3 doses unless gangrenous or perforated appendicitis)
- **Abscess:** ceftriaxone + metronidazole **OR** ciprofloxacin + metronidazole **OR** carbapenems **OR** Pip-Tazo **OR** antifungals if indicated
- **Acute Pancreatitis:** N/A, Non-infectious

Additional Key Points

- **Appendicitis:** Generally, one dose of abx pre-op is sufficient, no abx required post-op (exception: gangrenous or perforated appendicitis)
 - Treatment should have adequate coverage for gram negative and anaerobes